ALGEBRA & GRAPHS

Inequalities, Sequences & Proportion

Complete coverage of three key algebra topics.

Use these notes well and you will be able to:

- Representandsolvelinearinequalities;show on a number line.
- Solve compound inequalities and graphical inequalities in 2D (F 02).
- Find and use the nth term of linear, quadratic, and cubic sequences.
- Continue sequences and find term-to-term rules.
- Express and apply direct and inverse proportion (F 02).

If you follow the steps, you will succeed.

Inequalities — Symbols and Rules

SYMBOL	MEANING	NUMBER LINE MARK	EXAMPLE
$<$	Strictly less than	Open circle	$x < 3$
$>$	Strictly greater	Open circle	$x > -1$
$\leq$	Less than or equal	Filled circle	$x \leq 5$
$\geq$	Greater or equal	Filled circle	$x \geq 0$

KEY RULE: when you multiply or divide both sides by a NEGATIVE number, REVERSE the inequality sign.

Solve and show on number line: $3x - 4 > 8$

Add 4: $3x > 12$
Divide by 3: $x > 4$
Number line: open circle at 4, arrow pointing right
$x > 4$

Solve compound inequality: $-3 \leq 3x - 2 < 7$

Add 2 throughout: $-1 \leq 3x < 9$
Divide by 3: $-1/3 \leq x < 3$
Number line: filled circle at $-1/3$, open circle at 3
$-1/3 \leq x < 3$

Solve: $4 - 3x \geq 13$

Subtract 4: $-3x \geq 9$
Divide by -3 and REVERSE sign: $x \leq -3$
$x \leq -3$

Graphical Inequalities in Two Variables [F 02]

Draw the boundary line. Use a DASHED line for strict inequalities ($<$ or $>$) and a SOLID line for $\leq$ or $\geq$. Test a point (usually (0,0)) to determine which side to shade. Shade the UNWANTED region (unless the question says otherwise).

Show the region defined by: $y \geq 2x - 1$ and $y < 3$

Boundary 1: $y = 2x-1$ solid line ($\geq$). Shade BELOW.
Boundary 2: $y = 3$ dashed line ($<$). Shade ABOVE (unwanted).
Test (0,0): $0 \geq -1$ and $0 < 3$ (0,0) is in the wanted region.
Required region: above $y=2x-1$ (solid) and below $y=3$ (dashed)

Sequences

Arithmetic (Linear) Sequences — nth term = $dn + (a - d)$ or $an + b$

Find the nth term of: 5, 8, 11, 14, ...

Common difference $d = 3$. First term $a = 5$.
nth term $= 3n + (5 - 3) = 3n + 2$
Check: $T(1)=5$ $T(4)=14$
$T = 3n + 2$

Is 98 a term of the sequence $T = 4n - 2$?

Set $4n - 2 = 98$ $4n = 100$ $n = 25$
25 is a positive integer
Yes — 98 is the 25th term

Quadratic Sequences — second differences are constant nth term has n^2

Find the nth term of: 2, 5, 10, 17, 26, ...

First differences: 3, 5, 7, 9, ... (increasing)
Second differences: 2, 2, 2, ... (constant = 2) coefficient of $n^2 = 2/2 = 1$
Start with n^2: 1, 4, 9, 16, 25 — compare to sequence: remainder 1, 1, 1, 1, 1
nth term $= n^2 + 1$
$T = n^2 + 1$

Find the nth term of: 3, 12, 27, 48, 75, ...

Divide each term by n^2: $3/1, 12/4, 27/9, 48/16, 75/25 = 3, 3, 3, 3, 3$
All give 3 constant ratio to n^2
$T = 3n^2$

Cubic Sequences — third differences are constant nth term has n^3

Show that 1, 8, 27, 64, 125, ... is a cubic sequence

This is n^3: $1^3, 2^3, 3^3, 4^3, 5^3$
First differences: 7, 19, 37, 61 | Second: 12, 18, 24 | Third: 6, 6 constant
$T = n^3$ (third differences constant $= 6 = 3! \times$ coefficient of n^3)

SEQUENCE TYPE	CONSTANT DIFFERENCES	nth TERM FORM
Linear	1st differences	$an + b$
Quadratic	2nd differences	$an^2 + bn + c$
Cubic	3rd differences	$an^3 + bn^2 + cn + d$

Proportion

Proportion describes a multiplicative relationship between two variables.

TYPE	SYMBOL	EQUATION	GRAPH SHAPE
Direct: linear	$y \quad x$	$y = kx$	Straight line through origin
Direct: square	$y \quad x^2$	$y = kx^2$	Parabola through origin
Direct: square root	$y \quad \sqrt{x}$	$y = k\sqrt{x}$	Square root curve
Direct: cube	$y \quad x^3$	$y = kx^3$	Cubic through origin
Inverse: linear	$y \quad 1/x$	$y = k/x$	Reciprocal curve
Inverse: square	$y \quad 1/x^2$	$y = k/x^2$	Steeper reciprocal

Tip: k is the CONSTANT OF PROPORTIONALITY — find it by substituting a known pair of values. Once k is found, the equation is complete and you can find any unknown.

y is directly proportional to x^3. When $x = 2$, $y = 40$. Find y when $x = 3$.

$y \quad x^3 \quad y = kx^3$
Substitute: $40 = k(2^3) = 8k \quad k = 5$
Equation: $y = 5x^3$
When $x = 3$: $y = 5(27)$
$y = 135$

y is inversely proportional to $\sqrt{x}$. When $x = 9$, $y = 4$. Find x when $y = 12$.

$y \quad 1/\sqrt{x} \quad y = k/\sqrt{x}$
Substitute: $4 = k/\sqrt{9} = k/3 \quad k = 12$
Equation: $y = 12/\sqrt{x}$
When $y = 12$: $12 = 12/\sqrt{x} \quad \sqrt{x} = 1 \quad x = 1$
$x = 1$

Practice Questions

INEQUALITIES

1. Solve: $5x + 3 > 2x - 9$. Show on a number line.

2. Solve: $-2 < 4x + 2 \leq 14$. List the integer values.

3. Solve: $2(3x - 1) \leq 4x + 8$

4. List the inequalities defining the region where $x \geq 0$, $y \geq 1$, $x + y \leq 5$

SEQUENCES

5. Find the nth term of: 7, 11, 15, 19, ...

6. Find the nth term of: 0, 3, 10, 21, 36, ... (quadratic)

7. Is 200 a term in the sequence $T \ = 5n - 3$? Show working.

8. The 4th term of a linear sequence is 22 and the 7th term is 37. Find the nth term.

PROPORTION

9. p is inversely proportional to q^2. When $p = 4$, $q = 3$. Find p when $q = 6$. [F 02] 10. y is

directly proportional to x^2. When $x = 5$, $y = 75$. Find y when $x = 4$. [F 02]

ADD YOUR NOTES HERE

Hey smart student!

Sequences:lookatdifferences first — 1st, then 2nd, then 3rd.
Constant differences tell you the degree (linear, quadratic, cubic).

Proportion: find k FIRST using the given pair. Then write the equation.

Keep going — see you in the next lesson! From FREDi